Program - 10: Build and Test Applications with CI

Aim: To install and configure Jenkins to automatically build and execute a Java application and demonstrating the basic concept of Continuous Integration (CI).

Objectives

- To understand the workflow of Continuous Integration.
- To create a simple Java application using Eclipse.
- To initialize and manage the application using Git.
- To configure Jenkins for automated build execution.
- To compile and execute a Java program automatically using Jenkins.
- To verify the build status through the Jenkins console output.

Problem Statement

In traditional software development, developers manually compile and test applications when changes are made. This can be time consuming and may lead to errors.
The objective of this experiment is to use Jenkins to automate the build and execution process of a simple Java application. Whenever the Jenkins job is executed, it should automatically compile and run the application and display the result.

Description

Jenkins is an automation server used to implement Continuous Integration and Continuous Delivery practices.
In this experiment, a simple Java program is created using Eclipse. The source code is managed using Git. A Jenkins free style project is then created to compile and execute the Java program automatically.

Requirements

Software:

- JDK 17 or JDK 21
- Eclipse IDE
- Git
- Jenkins
- Web browser

Procedure

Step 1 – Create a Java Application

- Open Eclipse and create a simple Java project.
- Create a Java class named HelloDevOps.java:

```
public class HelloDevOps {
   public static void main(String[] args) {
      System.out.println("Hello from DevOps Lab!");
      System.out.println("Jenkins CI is working successfully.");
   }
}
```

Step 2 – Initialize Git

Open Command Prompt in the project directory and run:

- git init
- git add .
- git commit -m "Initial Java application"

Step 3 – Start Jenkins

- Start Jenkins and open the Jenkins dashboard at: http://localhost:8080

Step 4 – Create a Jenkins Project

1. Select: **New Item** \rightarrow **Freestyle Project**
2. Enter the project name: DevOps - HelloWorld
3. Click **OK**

Step 5 – Configure the Build

1. Go to: **Build** \rightarrow **Add build step** \rightarrow **Execute Windows batch command**
2. Enter the commands:
   javac HelloDevOps.java
   java HelloDevOps

3. Click **Save**

Step 6 – Execute the Jenkins Job

Click **Build Now**.
Jenkins will automatically:

1. Compile Java Program
2. Execute Java Program
3. Display Output
4. Show Build Status

Step 7 – View the Result

1. Click: **Build Number** \rightarrow **Console Output**
2. Check whether the build was successful.

Conclusion

CI using Jenkins automates the entire software build, test, and deployment process. It ensures faster feedback on code changes, reduces manual effort, and improves software reliability and development efficiency.